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Physics-informed machine learning for the evaluation of full-field residual stresses in a multi-pass dissimilar steel weld

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Highlights

- A validated 3D thermo-elasto-plastic FE model of a SUS304–Q235B weld provides high-fidelity training data.
- PINN predicts full-field residual stresses from geometry, thermal history, and von Mises constraints.
- PINN outperforms BPNN and PSO-BP models with higher accuracy and robust generalization.
- Including thermal-history features and physics constraints in PINN improves model performance.

Abstract

This paper proposes a physics-informed neural network (PINN) machine learning method guided by physical constraints to achieve rapid and high-precision evaluation of full-field residual stresses in a dissimilar SUS304-Q235B steel weld. To directly build an end-to-end prediction model from spatial information and thermal history characteristics to six-component stress fields, the model uses multi-dimensional inputs, including the three-dimensional coordinates of nodes and the key thermal history characteristics (peak pass temperature, maximum cooling rate and average cooling rate of the final cooling stage at each stage of the four-pass welding process), and the von Mises yield criterion is embedded in the loss function as a physical constraint. The reliability of the finite element model is firstly experimentally verified through the measured results by using contour method, and training data is then generated using the finite element results. Compared to a conventional backpropagation neural network and a particle-swarm-optimization-based neural network, the proposed PINN achieves substantial reductions in both root-mean-square error (RMSE) and mean absolute error (MAE), demonstrating its superior predictive capability, and it yields ~60% higher efficiency than the particle-swarm-optimization-based neural network. An ablation study further highlights the effectiveness and necessity of integrating physical information and imposing physical constraints to achieve accurate stress predictions.

Keywords

Dissimilar steel weld; Residual stress; Physics-informed neural network; Finite element modeling

CRediT authorship contribution statement

Declaration of competing interest

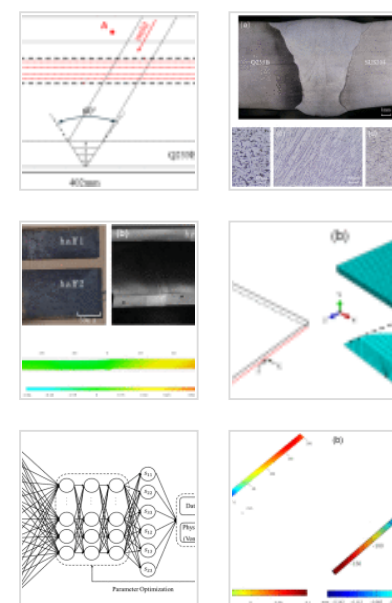
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1. Introduction

Table 3. Calculation time of each model.

Dissimilar steel welded joints are extensively employed in large-scale pressurized components, such as supercritical boilers and nuclear power pipelines [1,2], owing to their ability to combine the complementary advantages of different materials, including high-temperature strength, corrosion resistance, and cost efficiency. However, multi-pass welding inevitably induces substantial residual tensile stresses in the weld metal and heat-affected zones, which can significantly increase susceptibility to stress corrosion cracking, fatigue crack propagation, and brittle fracture [[3], [4], [5]]. Accurate characterization of the residual stress field is therefore essential for assessing the structural integrity and service reliability of these components under extreme operating conditions [6,7].

A variety of destructive and non-destructive experimental techniques have been developed to evaluate residual stress distributions in dissimilar steel welds [[7], [8], [9]]. For instance, neutron diffraction and the contour method have been successfully applied to measure through-thickness residual stress profiles in such joints [9,10]. Nevertheless, these experimental approaches are often constrained by limited spatial resolution, restricted accessibility, high cost, and low efficiency, making it difficult to obtain full-field residual stress distributions. In parallel, thermo–metallurgical–mechanical coupled finite element models have been widely employed to predict welding-induced residual stresses [[11], [12], [13], [14]]. Despite their proven accuracy, the complexity of model construction and the associated high computational cost limit their applicability for rapid prediction and real-time decision-making in engineering practice.

In recent years, machine-learning approaches based on neural networks have emerged as an efficient alternative for residual stress prediction in welding processes. By establishing nonlinear mappings between process parameters and residual stress responses, these methods significantly reduce the computational burden associated with conventional finite element simulations. Research in this area has followed a clear evolutionary trajectory, progressing from the application of basic neural network architectures [[15],

[16], [17]] to advanced hybrid intelligent frameworks that integrate multiple optimization algorithms and are tailored to specific engineering scenarios [[18], [19], [20], [21]]. For example, a backpropagation neural network optimized using the Levenberg–Marquardt algorithm demonstrated excellent predictive capability for gas metal arc welding of aluminum alloy T-joints, achieving a correlation coefficient of 0.983 on the testing dataset and prediction errors below 2.5% [15]. Studies by Mathew et al. [19,20] and Rissaki et al. [21] further advanced residual stress prediction for circumferential welds in austenitic stainless steel pipes, where neural network predictions were systematically compared with engineering structural integrity assessment criteria, facilitating the transition of data-driven approaches from academic research to engineering applications. To further mitigate overfitting and enhance predictive robustness, Park and Na [22] developed a deep fuzzy neural network incorporating rule-dropout mechanisms and genetic algorithms, achieving root-mean-square errors below 1.8% in residual stress prediction for dissimilar metal welds under nuclear power plant operating conditions.

The data-driven prediction methods of welding residual stress mainly develop along two technical paths: one relies on the infrastructure represented by back propagation neural network (BPNN) [23,24], whose performance constitutes the baseline of many current engineering prediction models, which is simple in structure but easy to fall into local optimization and sensitive to initial values; The other is to improve the performance in data-driven paradigm through intelligent optimization strategies (such as particle swarm optimization back-propagation neural network (such as PSO-BPNN)) [25], and optimize network parameters by meta-heuristic algorithm [26,27].

Despite these advances, most existing neural-network-based models remain essentially “black boxes” [28], lacking explicit incorporation of physical mechanisms. This deficiency often leads to overfitting and limited generalization capability, particularly when predictions are extrapolated beyond the boundaries of the training data, where errors may increase dramatically [29]. To address these limitations, physics-informed neural networks (PINNs) have recently emerged as a promising alternative [30]. The fundamental concept of PINNs is to embed governing physical equations or constraints

directly into the loss function of the neural network, thereby guiding the training process to satisfy physical laws while fitting available data. This paradigm has demonstrated considerable success across a wide range of scientific and engineering applications [[31], [32], [33]], particularly under data-scarce conditions, by enhancing both predictive accuracy and physical consistency [34,35].

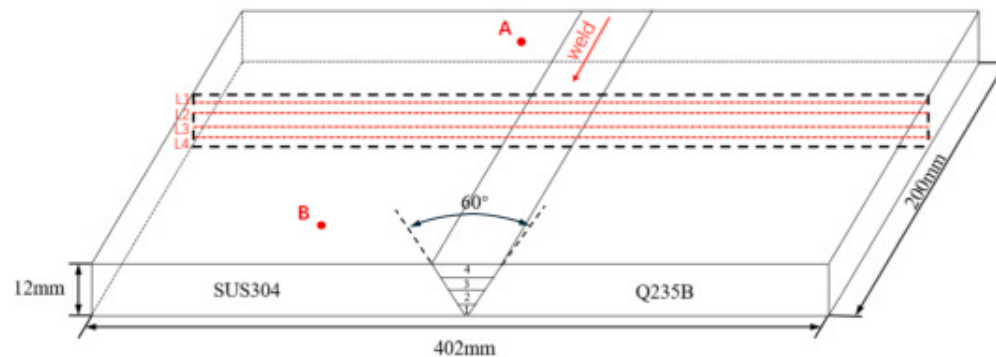
The formation and spatial distribution of welding residual stress are controlled by the elastic-plastic response of the material during thermal cycling, in which the yield criterion plays a decisive role. In this study, the von Mises yield criterion is embedded in the loss function of the neural network as a physical information constraint; at the same time, a set of multi-stage thermal features, namely the peak temperature, maximum cooling rate and average cooling rate of each weld bead, are introduced as key input features to comprehensively characterize the thermal history. This framework builds a residual stress prediction model that integrates data-driven learning and physics guidance. By embedding these physical constraints, the proposed model establishes an inherent inter-component coupling mechanism within the stress tensor, forcing the optimization process to consider the interdependencies between all stress components, thus improving the reliability of predictions.

The remainder of this paper is organized as follows. Section 2 describes the experimental procedures for fabricating the dissimilar steel welded joint and the subsequent residual stress measurements. Section 3 presents the numerical modeling strategy and the physics-informed neural network framework, wherein the experimental results are used to validate the sequentially coupled finite element model, which subsequently provides the database for neural network training, validation, and testing. Section 4 discusses the results in detail, and Section 5 summarizes the main conclusions of this study.

2. Experiment

2.1. Fabrication of the dissimilar steel weld

Fig. 1 shows the geometric configuration of the welded specimen. Manual arc welding was employed to fabricate a dissimilar joint between SUS304 austenitic stainless steel and Q235B low-carbon steel using an E309-16 electrode with a diameter of 4.0mm on a Beijing Times PE50-500 welding machine. The welding procedure consisted of four layers and four passes, and the key process parameters are summarized in Table 1. During welding, the interlayer temperature was strictly maintained below 150°C. A tack weld was performed prior to welding to ensure proper assembly and alignment. To record real-time thermal cycles during welding, two K-type thermocouples were spot-welded onto the plate surface at the locations indicated in Fig. 1. Point A is positioned 10mm from the weld line and 10mm from the starting end of the weld, whereas Point B is located 55mm from the weld edge and 10mm from the terminal end of the weld.



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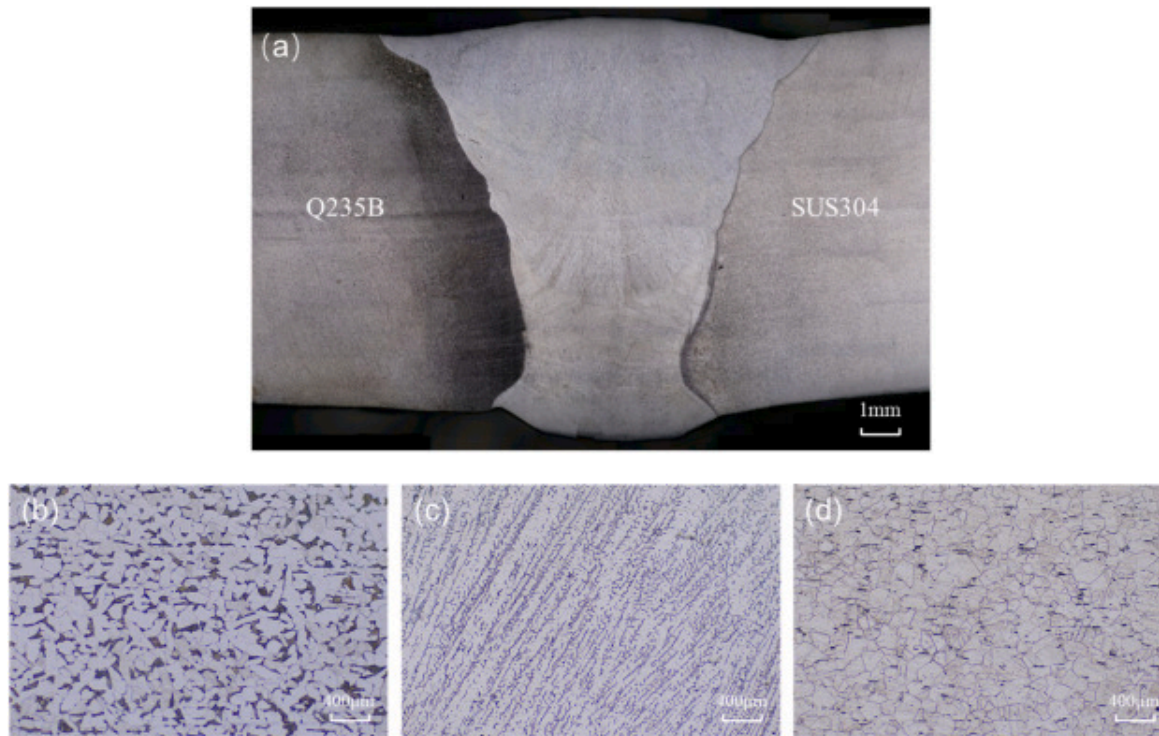
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Fig. 1. Geometry of the welded specimen and locations of temperature measurement points.

Table 1. Welding process parameters used in the fabrication of the dissimilar joint.

Voltage (V)	Current (A)	Speed (mm/s)
20–25	135–145	20–30

In order to systematically evaluate the forming quality and microstructure characteristics of the welded joint, macro and micro metallographic analysis of the weld cross section were carried out. As shown in Fig. 2 (a), the macro metallographic photos clearly show that the weld bead is complete, complete penetration is achieved, and there are no visible cracks, non-fusion and other macro defects. The weld metal (WM) and the base metal on both sides (Q235B and SUS304) and HAZ are clearly defined.



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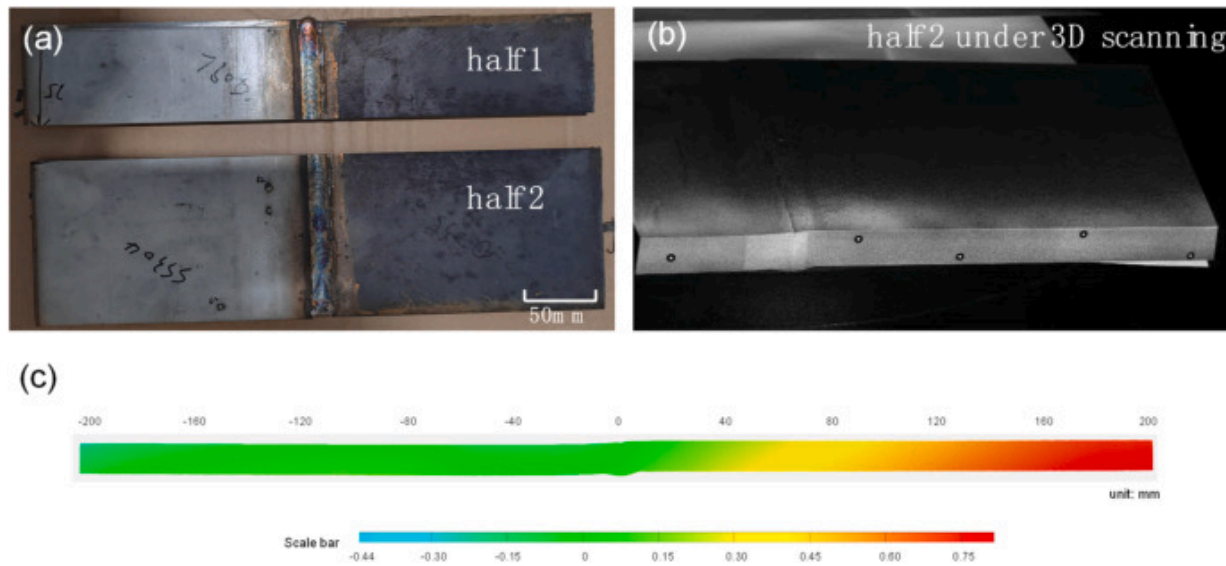
Fig. 2. Macroscopic and microscopic characterization of the dissimilar welded joint.

Further microscopic observation is carried out under an optical microscope. Typical structures of each region are shown in Fig. 2(b–d). The heat affected zone on the Q235B side (Fig. 2(b)) presents a mixed structure of ferrite and pearlite; the heat affected zone

on the SUS304 side (Fig. 2(c)) maintains austenite matrix, with clear grain boundaries, and carbide precipitates along grain boundaries in some regions; The weld zone of E309-16 (Fig. 2(d)) shows austenite dendrite structure formed by solidification of welding material. These macro and micro structural characteristics together confirm the reliability of welding process.

2.2. Contour method measurement of residual stresses

The contour method, originally developed by Prime [36], was employed to experimentally determine the residual stress distribution in the dissimilar steel weld. After welding, the as-welded specimen was sectioned into two halves using wire electrical discharge machining (EDM) with a 200- μm -diameter wire (see Fig. 3(a)). During cutting, the specimen was symmetrically clamped and submerged in deionized water, and the cut was initiated 75 mm from the weld start position. Subsequently, following the procedure described in Ref. [37], an ATOS Q 3D optical scanner (GOM GmbH) was used to measure the stress-relief-induced deformation of the exposed cross-section. Fig. 3(b) shows the specimen half 2 under 3D scanning, and Fig. 3(c) presents the obtained contour plot of the deviation displacement from the cut surface after the preliminary data processing on the point cloud data that removed the non-useful geometric information other than the target cross-section surface.



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Fig. 3. Schematic illustration of the contour method procedure: (a) specimen after wire EDM sectioning, (b) dissimilar steel weld during optical scanning, and (c) contour plot of the deviation displacement from the cut surface.

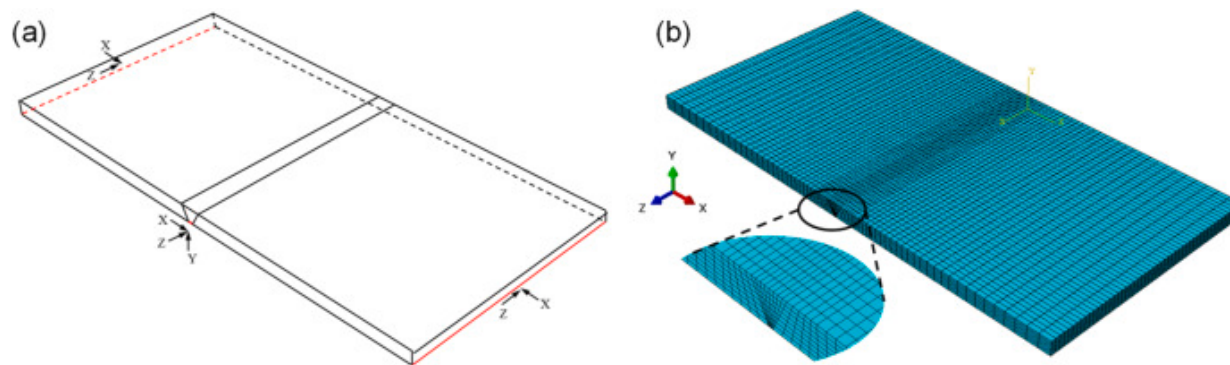
An in-house data processing tool [37] was employed to generate spline-fitted contours on the cross-sectional surface, which were subsequently converted into boundary conditions for a linear elastic finite element model to calculate the residual stresses in the weld. The data-processing procedure is summarized as follows: using a Python script, interpolation points were sampled at 0.5 mm intervals along the x-direction and 0.1 mm along the y-direction; the X and Y coordinate systems of the two cut surfaces were unified to ensure accurate correspondence between measurement points; the deformation values at each point were averaged, denoised, and flattened. Based on these processed data, the contour curves were fitted using a knot spacing of 6 mm, and the resulting boundary conditions were extracted for finite element analysis. In the linear elastic simulations, the materials were assigned a Young's modulus of 210 GPa and a Poisson's ratio of 0.3, and the 20-node

reduced-integration continuum element C3D20R was used to generate the mesh. Further details on the implementation process of contour method can be found in Refs. [36,37].

3. Modeling and prediction approach

3.1. Finite element model

The training of physical information neural networks (PINN) relies on high quality label data. To this end, an experimentally validated sequential coupled thermo-elastic-plastic finite element model is constructed with the sole purpose of generating reliable data sets for subsequent PINN training and testing. As illustrated in Fig. 4, a three-dimensional (3D) finite element model was constructed based on the weld specimen dimensions. The mesh was refined within the weld zone and coarsened in the surrounding regions, with an average mesh size consistent with that used in the contour method. Heat transfer analysis was performed using 8-node hexahedral DC3D8 elements, while mechanical analysis employed reduced-integration C3D8R continuum elements. The model comprised a total of 33,507 elements and 38,800 nodes, with the smallest element size of ~ 2 mm located in the weld zone (see Fig. 4(b)).



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Fig. 4. Schematic of the finite element model: (a) geometric model and boundary conditions, and (b) finite element mesh.

A sequentially coupled thermo-mechanical approach was implemented to predict both temperature evolution and the full-field residual stresses in the dissimilar steel weld. To simulate the multi-pass deposition process, the “element birth and death” technique was used to progressively activate the weld elements layer by layer. In the thermal analysis, the double-ellipsoid heat source model proposed by Goldak [38] (Eq. (1)) was adopted and implemented via a user-defined subroutine in Abaqus.

$$q_f(x, y, z) = \frac{6\sqrt{3}f_fQ}{\pi\sqrt{\pi}a_fb c} \exp\left[-3\left(\frac{x^2}{a_f^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right]$$

$$q_r(x, y, z) = \frac{6\sqrt{3}f_rQ}{\pi\sqrt{\pi}a_r b c} \exp\left[-3\left(\frac{x^2}{a_r^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right] \quad (1)$$

In Eq. (1), $Q = \eta UI$ is the heat input power (W); a_f , a_r are the semi-axis parameters of the front and rear hemispheres (mm); b is the weld width (mm); c is the weld penetration depth (mm); and f_f , f_r are the energy distribution coefficients in the front and rear hemispheres, with $f_f=0.7$ and $f_r=1.3$. The parameters of the heat source model were calibrated using the experimentally-measured thermal cycles during the welding process and listed in Table 2.

Table 2. Calibrated parameters of the double-ellipsoid heat source model (unit: mm).

a_f	a_r	b	c
9	15	20	5

To replicate realistic heat transfer conditions, a convective heat transfer coefficient of $15\text{Wm}^{-2}\text{K}^{-1}$ and a thermal radiation emissivity of 0.85 were applied to the outer surfaces of the model [39]. The resulting thermal fields were then used as input for the subsequent mechanical analysis. In the mechanical simulations, both the base metals and weld metal

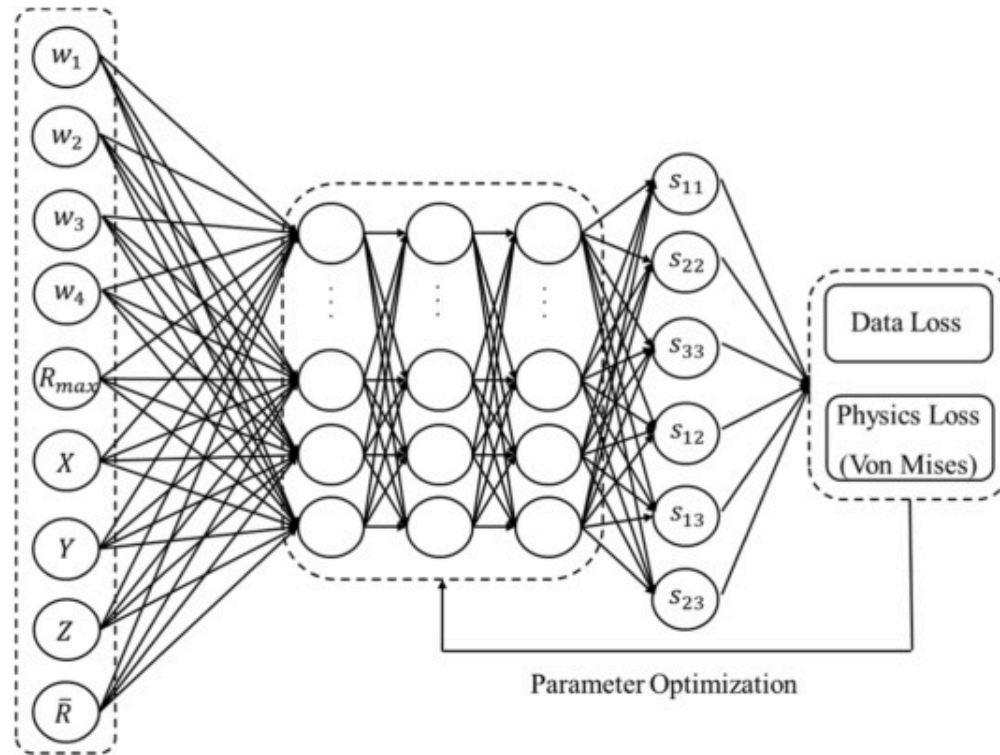
were assumed to exhibit isotropic elastic–plastic behavior. To prevent rigid body motion, the degrees of freedom in the x and z-directions were constrained for all nodes along the edges of the bottom plate and at the start of the weld, as shown in Fig. 4(a). Additional temperature-dependent thermo-physical and mechanical properties required for the coupled analysis, including thermal conductivity, density, Young's modulus, Poisson's ratio, and coefficient of thermal expansion, can be found in Refs. [[40], [41], [42]].

3.2. Physics-constraint guided neural network

The first step in building a neural network is to select input features that are closely related to the target residual stress field. The input feature design in this paper is guided by the physical mechanism of welding thermal-mechanical coupling. The physical laws are used as structured prior knowledge and deeply embedded in the architecture and learning objectives of the neural network to build a 'physical knowledge-enhanced agent model' with physical laws as the skeleton and data as the muscle. This paradigm aims to fundamentally improve the prediction rationality and generalization ability of the model in data-scarce areas, transforming the neural network from a simple data fitter to an efficient solver of field variables under physical constraints.

Concrete implementation begins with physically meaningful feature engineering. The selection of input features directly corresponds to the key physical dimensions driving the evolution of residual stresses. The schematic framework of the proposed PINN is presented in Fig. 5. To fully capture the multi-stage nature of the welding process, the thermal history is encoded through the peak temperatures of each welding pass (w_1, w_2, w_3, w_4), which represent the spatial and temporal distribution of the heat input. The maximum cooling rate ($R_{\max}$) serves as a core indicator of the transient thermal gradient, directly linked to the instantaneous thermal strain rate in thermo-elastoplastic theory and acting as a primary driver of thermal stress. Furthermore, the average cooling rate during the final cooling stage (R) is introduced to characterize the overall cooling intensity, which influences the final stress state and, in materials such as dissimilar steels, may relate to phase transformation kinetics. For material systems prone to solid-state

phase transformation, the combination of $R_{\max}$ and R provides critical insight into the cooling path and the potential for transformation-induced volume strain—a significant contributor to residual stress. Collectively, this seven-dimensional input feature set ($w_1, w_2, w_3, w_4, R_{\max}, R$) offers a compact yet physically comprehensive representation of the thermal-mechanical coupling inherent to multi-pass welding.



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Fig. 5. Schematic of the proposed physics-informed neural network.

As a scalar strength criterion that describes the yield behavior of materials under multi-axial stress states, the von Mises criterion has a simple form, universal physical meaning and efficient calculation compared to directly introducing complex and multi-field coupled welding transient control equations. Although it does not directly describe the

evolution process, it can effectively capture the intrinsic coupling relationship that must be satisfied between the components of the stress tensor. Using the prediction error of von Mises stress as the loss term L_{VM} is equivalent to applying a “physically reasonable” regularization to the six-dimensional stress output space of the network. In backpropagation, the network parameters are simultaneously guided by the data fitting error and the physical constraint error gradient, which forces the mapping function learned by the network to not only approximate the data sample, but also tend to generate a stress field that satisfies the basic material strength theory, thus effectively suppressing the non-physical solution that a pure data-driven model may produce outside the training area.

To eliminate dimensional differences and improve network convergence stability, both the input features and target stress components were subjected to min–max normalization, mapping values to the range $[-1, 1]$:

$$x' = 2 \frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1 \quad (2)$$

where x denotes the original value, while $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the corresponding feature in the training set, respectively.

The maximum cooling rate, $R_{\max}$, is a critical input feature derived from the original temperature–time series. For each discrete point within a cooling segment, the instantaneous cooling rate was calculated using an unequally spaced three-point central difference method:

$$R(t_i) = - \frac{T(t_{i+1}) - T(t_{i-1}))}{t_{i+1} - t_{i-1}} \quad (3)$$

where $T(t_i)$ is the temperature at time t_i , and t_i is the absolute time in seconds. The maximum value among all $R(t_i)$ within the cooling stage was taken as $R_{\max}$ for that node. The average cooling rate during the final cooling stage, denoted R , was calculated as

the linear cooling rate over that specific phase:

$$R = \frac{T_{start} - T_{end}}{t_{start} - t_{end}} \quad (4)$$

where T_{start} and T_{end} are the temperatures at the beginning and the end of the final cooling stage, respectively, and t_{start} and t_{end} are the corresponding times. R_{max} captures the most severe thermal transient, while R represents the overall cooling intensity of the post-weld period.

To incorporate physical plausibility into the predicted stress field, a physics-based loss term was introduced. For the stress tensor at any node, the von Mises equivalent stress is defined as:

$$\sigma_{VM} = \sqrt{\frac{1}{2} \left[(s_{11} - s_{22})^2 + (s_{22} - s_{33})^2 + (s_{33} - s_{11})^2 \right] + 3(s_{12}^2 + s_{13}^2 + s_{23}^2)} \quad (5)$$

where $s_{11}, s_{22}, s_{33}, s_{12}, s_{13}, s_{23}$ are the six components of the stress tensor. During network training, the error between the predicted and true von Mises stress values is jointly optimized with the data-driven component, ensuring that the network not only fits all six stress components but also maintains overall physical consistency:

$$L_{VM} = \frac{1}{N} \sum_{i=1}^N \left| \sigma_{VM}^{pred,i} - \sigma_{VM}^{true,i} \right| \quad (6)$$

where $\sigma_{VM}^{pred,i}$ and $\sigma_{VM}^{true,i}$ are the predicted and reference von Mises stresses for node i , respectively. To further improve robustness, a relative error term was introduced to penalize deviations in high-stress regions, preventing the model from overfitting low-stress areas:

$$L_{rel} = \frac{1}{N} \sum_{i=1}^N \frac{|\sigma_{VM}^{pred,i} - \sigma_{VM}^{true,i}|}{|\sigma_{VM}^{true,i}| + \varepsilon} \quad (7)$$

where ε is a small constant to avoid division by zero.

The total loss function is composed of a primary data-driven term and physics-informed constraints. The data-driven term uses the mean absolute error (MAE) of the stress

components to balance contributions from all components and reduce sensitivity to outliers:

$$L_{MAE} = \frac{1}{6N} \sum_{i=1}^N \sum_{j=1}^6 \left| \sigma_{ij}^{pred,i} - \sigma_{ij}^{true,i} \right| \quad (8)$$

The overall loss function is then defined as:

$$L_{total} = L_{MAE} + \lambda_{VM} L_{VM} + \lambda_{rel} L_{rel} \quad (9)$$

where, λ_{VM} and λ_{rel} are weighting coefficients that balance the contributions of data fitting and physical constraints. In this study, the optimal values of these coefficients (0.7 for λ_{VM} and 0.5 for λ_{rel}) were determined via grid search, allowing for a balanced and physically consistent model.

To systematically evaluate the performance of the proposed PINN model, a comparative analysis was conducted against traditional backpropagation (BP) neural networks and particle swarm optimization-enhanced BP (PSO-BP) neural networks. Additionally, ablation studies were performed to assess the contributions of key design elements in the model.

The dataset was randomly and uniformly partitioned into training, testing, and validation sets at a fixed ratio of 6:3:1 to ensure the rigor of the comparative analysis. Optimal hyperparameter combinations for each model were determined via independent grid searches, covering network depth (2–5 layers), number of neurons per layer (16, 32, 64, 128, 256, 512), and learning rates ranging from 1×10^{-5} to 1×10^{-3} . The BP neural network (BPNN) employed a four-layer fully connected hidden architecture (128–64–32–16) with ReLU activation, trained using the AdamW optimizer with a learning rate of 0.0003, an initial weight decay of 0.00001, and a batch size of 32. The PSO-BP network utilized a four-layer fully connected hidden architecture (256–128–64–32) with ReLU activation and was trained in two stages: first, a PSO stage with a swarm size of 200 particles over 250 iterations, an initial inertia weight of 0.9, cognitive (c_1) and social (c_2) coefficients of 2.0 and 0.5, and a velocity limit of 0.3; followed by fine-tuning using AdamW with a learning

rate of 0.0001 and a batch size of 32. The PINN adopted a four-layer hidden architecture (512–256–128–64) with the GELU activation function, trained using the AdamW optimizer with a learning rate of 0.0001, an initial weight decay of 0.001, and a batch size of 32. These individually optimized configurations were designed to evaluate the respective performance ceilings of each model and to isolate the contribution of physics-informed constraints to predictive accuracy.

To comprehensively assess the accuracy and robustness of the constructed PINN in predicting welding residual stresses, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) were used as evaluation metrics. RMSE and MAE quantify the magnitude of deviations between predicted and true values, while R^2 reflects the proportion of variance in the data explained by the model. The formulas for these metrics are defined as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{pred,i} - y_{true,i})^2} \quad (10)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_{pred,i} - y_{true,i}| \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_{pred,i} - y_{true,i})^2}{\sum_{i=1}^N (y_{true,i} - \bar{y}_{true})^2} \quad (12)$$

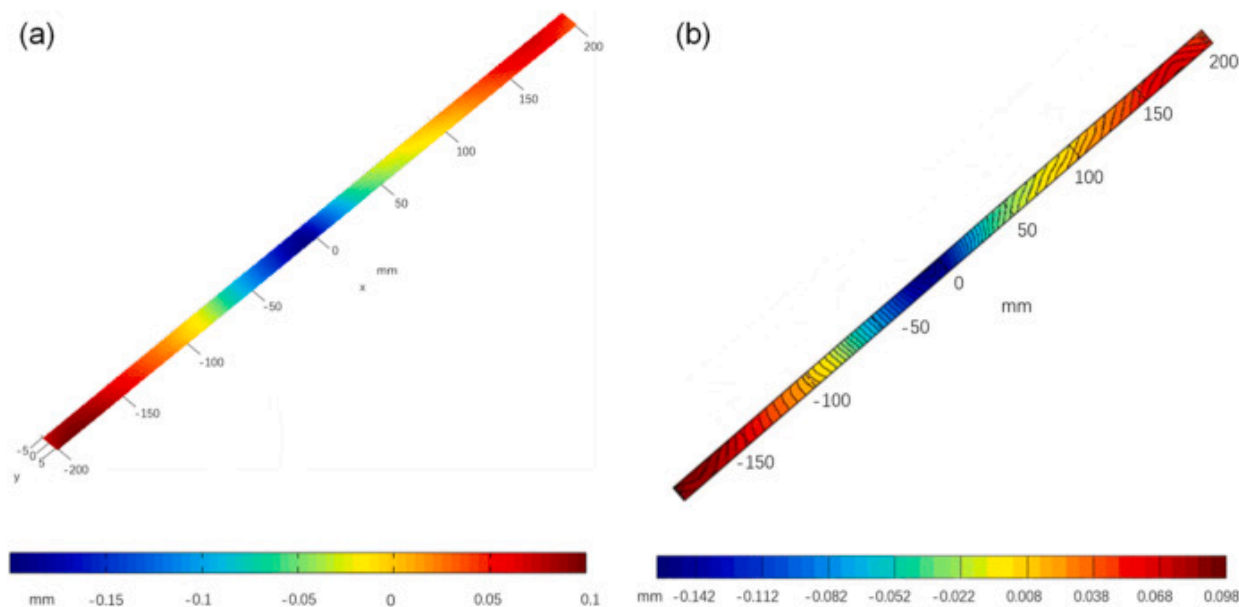
where $y_{pred,i}$ is the stress value predicted by the model, $y_{true,i}$ is the corresponding simulated stress value, and $\bar{y}_{true}$ is the mean of the simulated values.

4. Results and discussion

4.1. Measured residual stress distribution in the weld

As described in Section 2.2, a 3D scanner was used to perform full-field measurements on the two cut surfaces, acquiring displacement data along the longitudinal direction (normal to the cross-sectional surface). After smoothing and noise removal, the cross-sectional profiles and fitted curves were obtained, as shown in Fig. 6. The displacement

distribution indicates that stress-relief-induced deformations are negative within the weld zone, reaching a maximum of approximately $-160\mu\text{m}$, and gradually transition to positive values in the base metal, with a maximum of around $+100\mu\text{m}$. This pattern reflects that tensile residual stresses are concentrated in the weld zone and compressive stresses in the adjacent heat affected zone and base metal. Using the contour data from the cut surfaces, the longitudinal residual stress field was then inversely calculated in Abaqus, as discussed below.



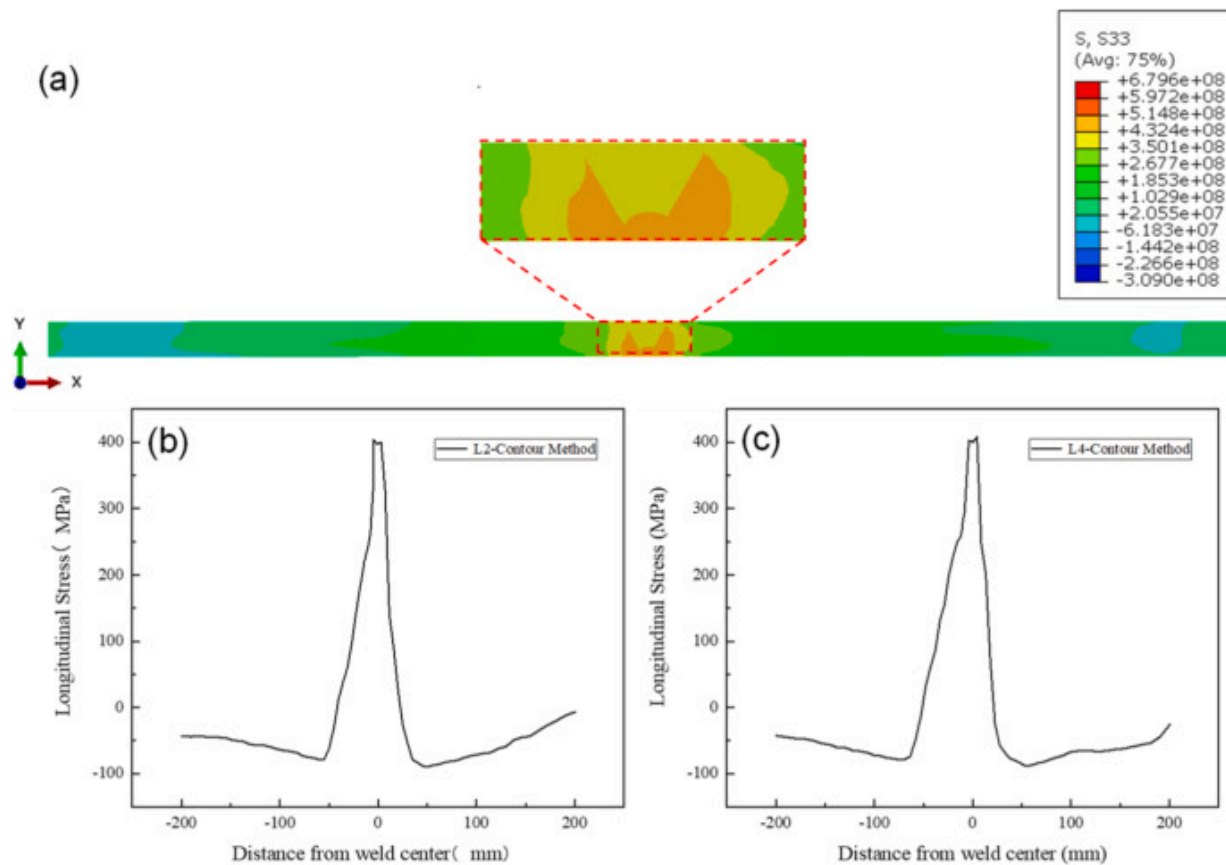
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Fig. 6. Measured out-of-plane displacements obtained by the 3D scanner: (a) contour plot and (b) spline-fitted results using a knot spacing of 6mm.

As shown in Fig. 7(a), the longitudinal residual stress of the welded joint shows a typical “tension-compression” self-equilibrium distribution, with the peak value of tensile stress in the center of the weld and the valley value of compressive stress in the heat-affected zone. The quantitative distribution of paths L2 and L4 along the plate thickness direction

(Fig. 7(b) & (c)) further shows that the peak tensile stress reaches 403MPa and 408MPa respectively, and the peak compressive stress is about -89MPa, and the peak value of the two paths is only about 5MPa different, indicating that the stress distribution in the thickness direction is uniform; the high tensile stress zone is concentrated in the melt width range, and the boundary stress in the heat-affected zone quickly drops below 100MPa, while the base metal zone tends to a low stress plateau.



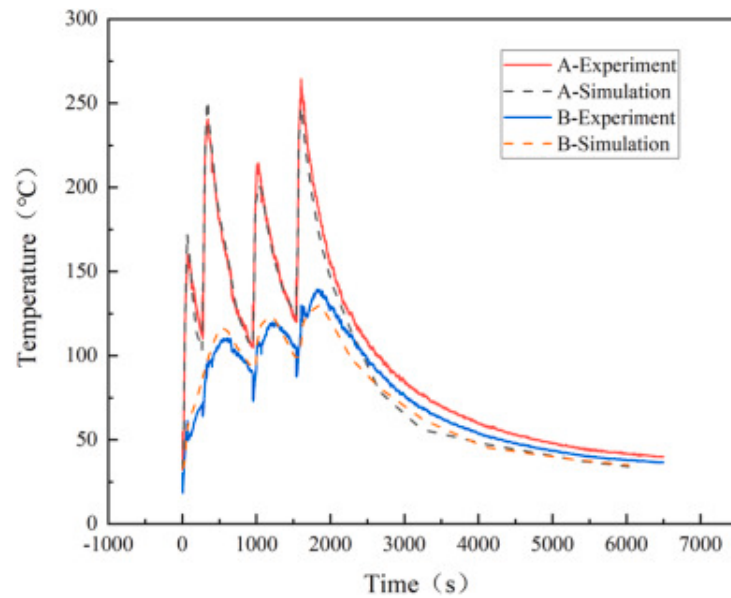
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Fig. 7. Longitudinal residual stress distribution in the weld: (a) cross-sectional contour, (b) along Path L2, and (c) along Path L4.

4.2. Validation of the finite element model

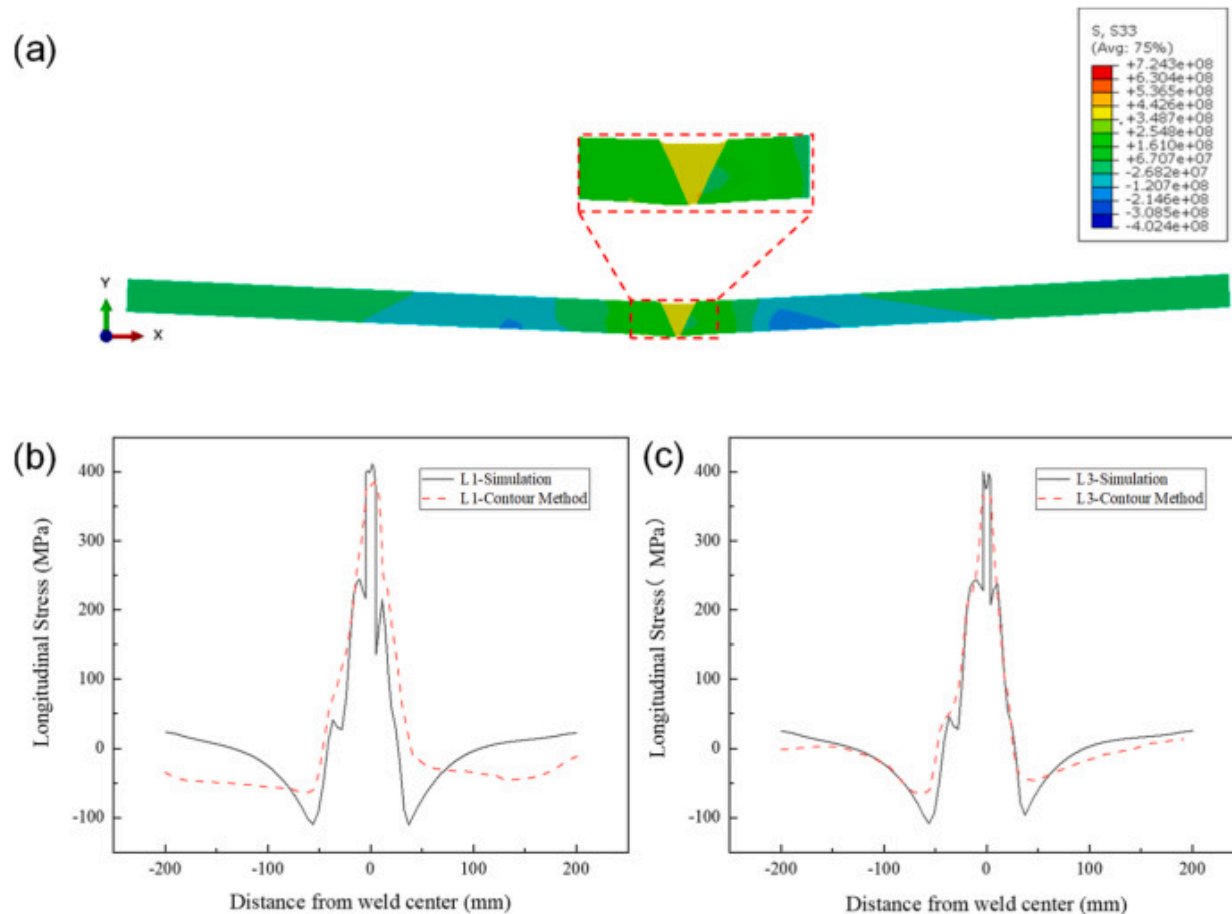
[Fig. 8](#) presents the thermal cycle curves for Point A (see [Fig. 1](#)), located near the weld, and Point B (see [Fig. 1](#)), situated in the far-weld region. The curve at Point A exhibits the characteristic rapid heating followed by exponential cooling typical of the welding process. The simulated thermal cycle closely matches the experimental measurements in terms of peak temperature and overall trend. Notably, during the cooling stage, the two curves almost overlap, indicating that the double-ellipsoidal heat source model and the thermophysical parameters used in the finite element model accurately capture the high heat input and subsequent dissipation in this region. At Point B, the peak temperature is substantially lower, and the simulated and experimental curves remain consistent in heating rate, peak temperature, and cooling behavior. These results confirm the reliability of the finite element model in reproducing the transient welding temperature field, providing a robust basis for accurate residual stress prediction. [Fig. 9](#) shows the longitudinal residual stress contour along the cross-section corresponding to the contour method experimental cut. The simulation reveals the characteristic “tension–compression” self-equilibrating stress pattern typical of welded joints.



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Fig. 8. Comparison between measured and predicted thermal cycles.



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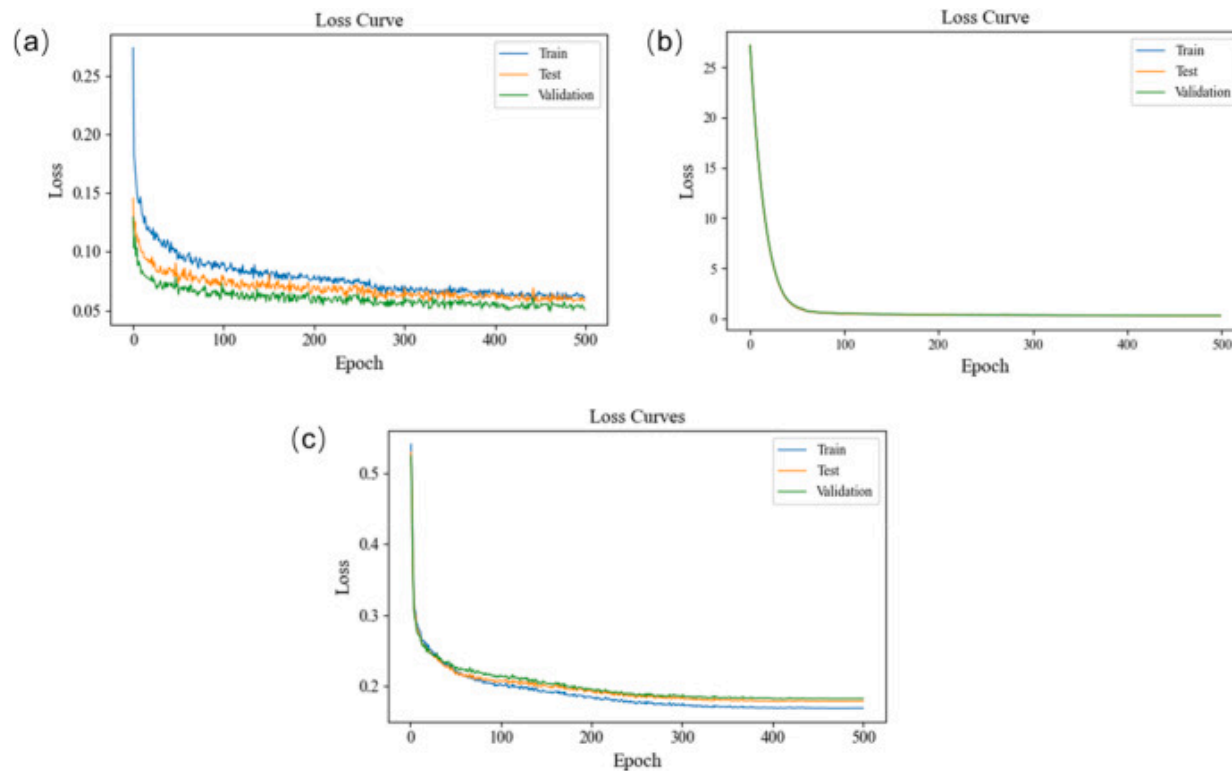
Fig. 9. Finite element predictions on longitudinal residual stresses: (a) 2D mapping on the cross-section of the weld at half-length and its comparison with contour method along (b) path L1 and (c) path L3.

For quantitative evaluation, the longitudinal residual stress distribution of the two paths (L1 and L3) corresponding to the profile method section was extracted from the section in Fig. 9(a), and compared with the experimental data obtained by the inverse operation of the profile method in Fig. 9(b) and (c). The tensile stress peak appears at the center of the

weld on both paths. The simulated and experimental values are 420MPa and 400MPa respectively in the L1 path, and 410MPa and 390MPa respectively in the L3 path. The relative error is controlled within 5%. The validity and accuracy of the established finite element model are verified, indicating that the data generated are suitable for subsequent training and testing of physical information neural networks.

4.3. Prediction performance of the proposed PINNs

Fig. 10 shows the training loss curves of three neural network models. The BP neural network (Fig. 10(a)) shows stable and consistent loss reduction on the training, test and validation sets, and the three curves converge completely simultaneously. Although its final loss value is relatively high (training set: 0.07271, test set: 0.06299, validation set: 0.05349), the test set and validation set losses are both lower than the training set, and the validation set loss is the lowest among the three, indicating that the model has good generalization ability. The PSO-BP network (Fig. 10(b)) uses particle swarm optimization to achieve faster initial convergence, and the overall loss is maintained at a low level (training set: 0.0263, test set: 0.0305, validation set: 0.0417). Although the optimization accelerates convergence, its generalization performance improvement is limited. In contrast, the proposed PINN (Fig. 10(c)) exhibits the most consistent and synchronized loss curve throughout the training process. Although its final loss value is higher (training set: 0.1691, test set: 0.1791, validation set: 0.1825), this is mainly due to the introduction of a physical loss term to make the model solution conform to physical laws. , but the three values are very close, indicating that the model has excellent stability and consistency on different data sets. This balanced convergence characteristic reflects PINN's strong generalization ability and reliable performance in practical applications.



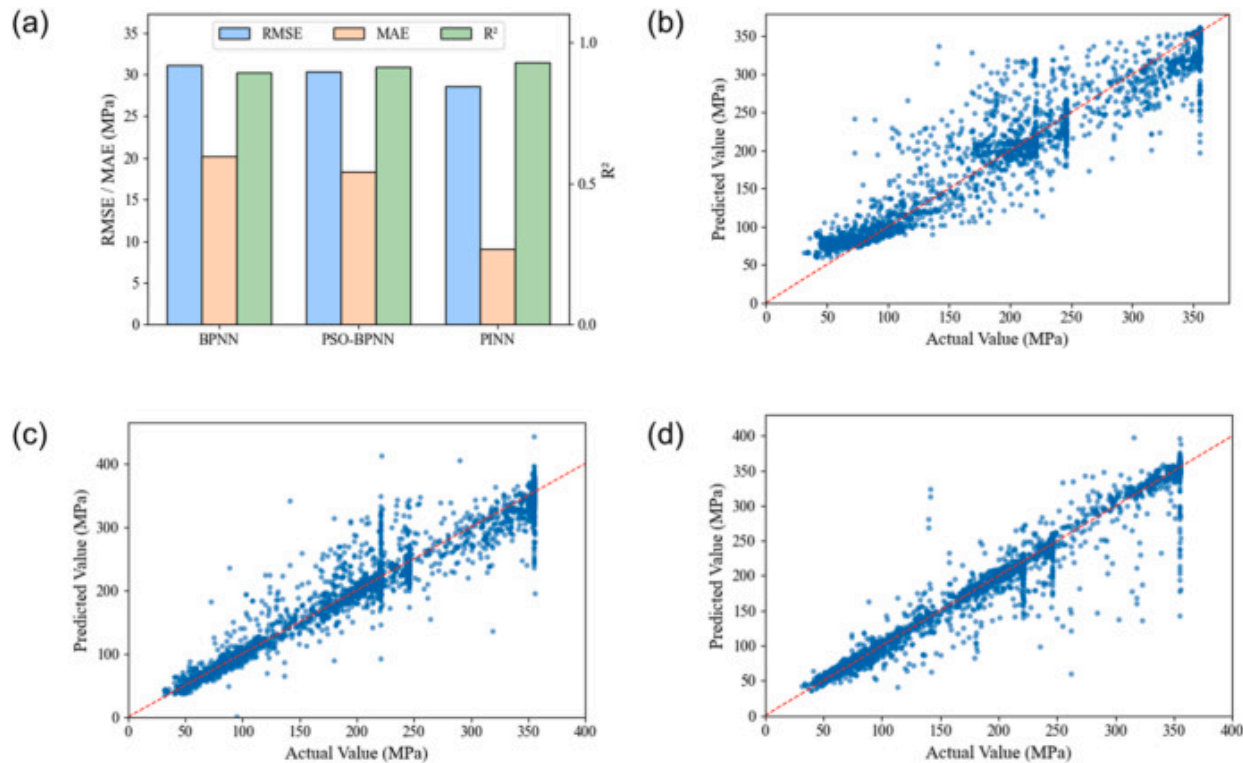
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Fig. 10. Evolution of training, validation, and testing losses with iterations: (a) BPNN, (b) PSO-BPNN, and (c) proposed PINNs.

To quantitatively assess model prediction accuracy, Fig. 11(a) compares performance metrics using destandardized stress values. The PINN model outperforms the other networks in several key metrics, achieving an RMSE of 28.6MPa and an MAE of 9.06MPa, both lower than those of the BPNN (RMSE=31.07MPa, MAE=20.17MPa) and PSO-BP (RMSE=30.37MPa, MAE=18.33MPa) models. Its coefficient of determination (R^2) reaches 0.9279, showing the model's strong predictive performance and reliability. Fig. 11(b)–11(d) further validate these findings. The scatter plots for BPNN and PSO-BP exhibit relatively dispersed distributions with noticeable deviations from the reference diagonal,

particularly in the medium stress range. In contrast, the PINN predictions are more closely aligned along the diagonal, demonstrating higher consistency across the entire stress spectrum and superior predictive capability relative to the other two models.



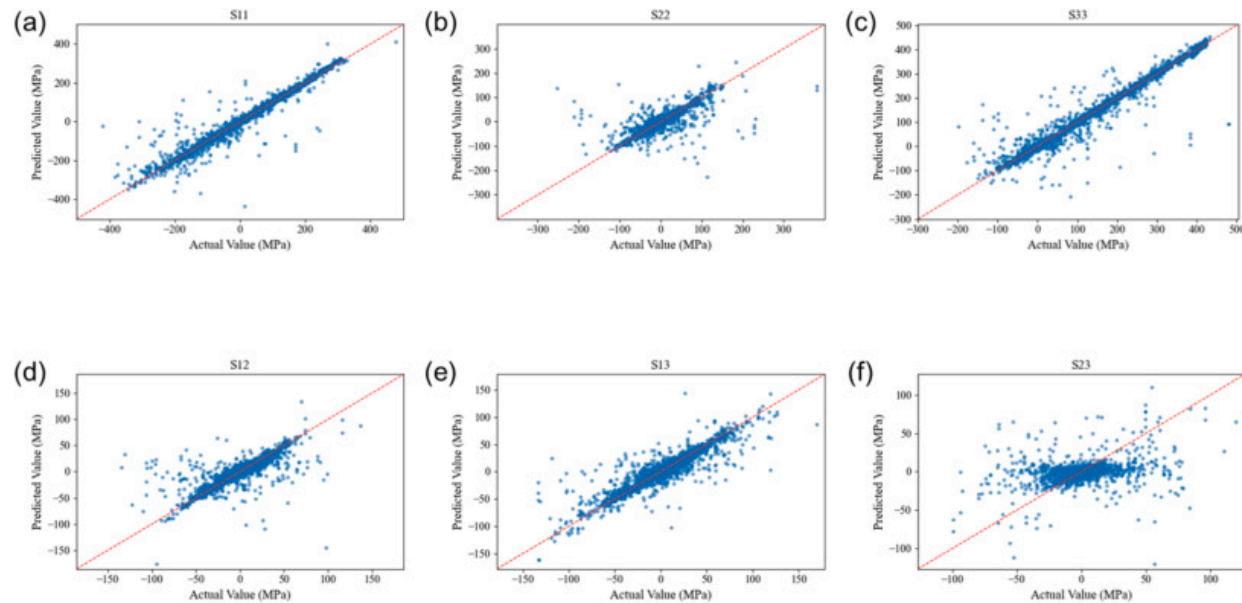
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Fig. 11. Prediction performance of different models: (a) evaluation metrics, (b) BPNN, (c) PSO-BPNN, and (d) proposed PINNs.

An individual evaluation of the six stress components predicted by the PINN (Fig. 12) reveals distinct levels of predictive performance. The R^2 values for five components (S_{11} , S_{33} , S_{13} , S_{12} , S_{22}) range between 0.4041 and 0.9555, indicating varying degrees of correlation with the actual values. Notably, S_{11} and S_{33} achieve high R^2 values of 0.9318 and 0.9555 respectively, while S_{12} and S_{13} also demonstrate moderate-to-strong

explanatory power with R^2 values of 0.7026 and 0.7931. In contrast, the S23 component exhibits poor correlation, with an R^2 of only 0.0671. These results demonstrate that the PINN model is capable of effectively reconstructing key stress components in the multi-axial residual stress field, although performance varies across different stress directions. The descaled RMSE and MAE metrics further support these findings, with S11 and S33 showing the highest predictive accuracy in engineering units (RMSE is 34.67MPa and 32.48MPa, MAE is 13.01MPa and 13.00MPa, respectively).



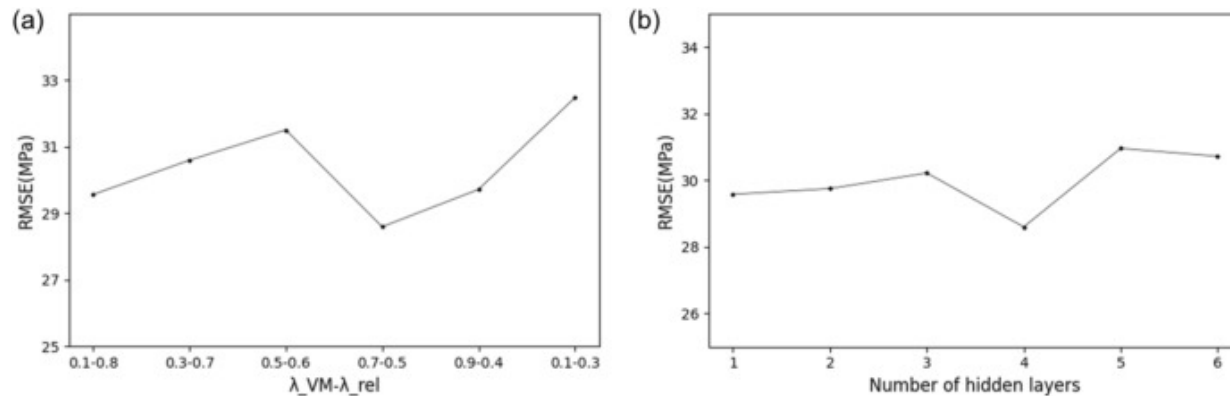
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Fig. 12. Prediction performance of individual stress components using the proposed PINN model.

In order to quantitatively evaluate the impact of physical loss weights (λ_{VM} , λ_{rel}) on prediction performance, a systematic parameter analysis was performed near the optimal combination determined by grid search. As shown in Fig. 13(a), as λ_{VM} increases, RMSE first decreases and then increases. When $\lambda_{VM}=0.7$ and $\lambda_{rel}=0.5$, the model achieves

optimal performance (RMSE=28.6MPa), at which point a balance is reached between physical constraints and data fitting. If λ_{VM} is too low (such as 0.1–0.3), the physical constraints are insufficient, and the model degenerates into an approximately pure data-driven model, resulting in an increase in MAE; if λ_{VM} is too high (such as 0.9), the physical constraints are too strong, inhibiting data learning, and also causing performance degradation. The performance plateau near the optimal parameters is wide, indicating that the model is robust to hyperparameters.



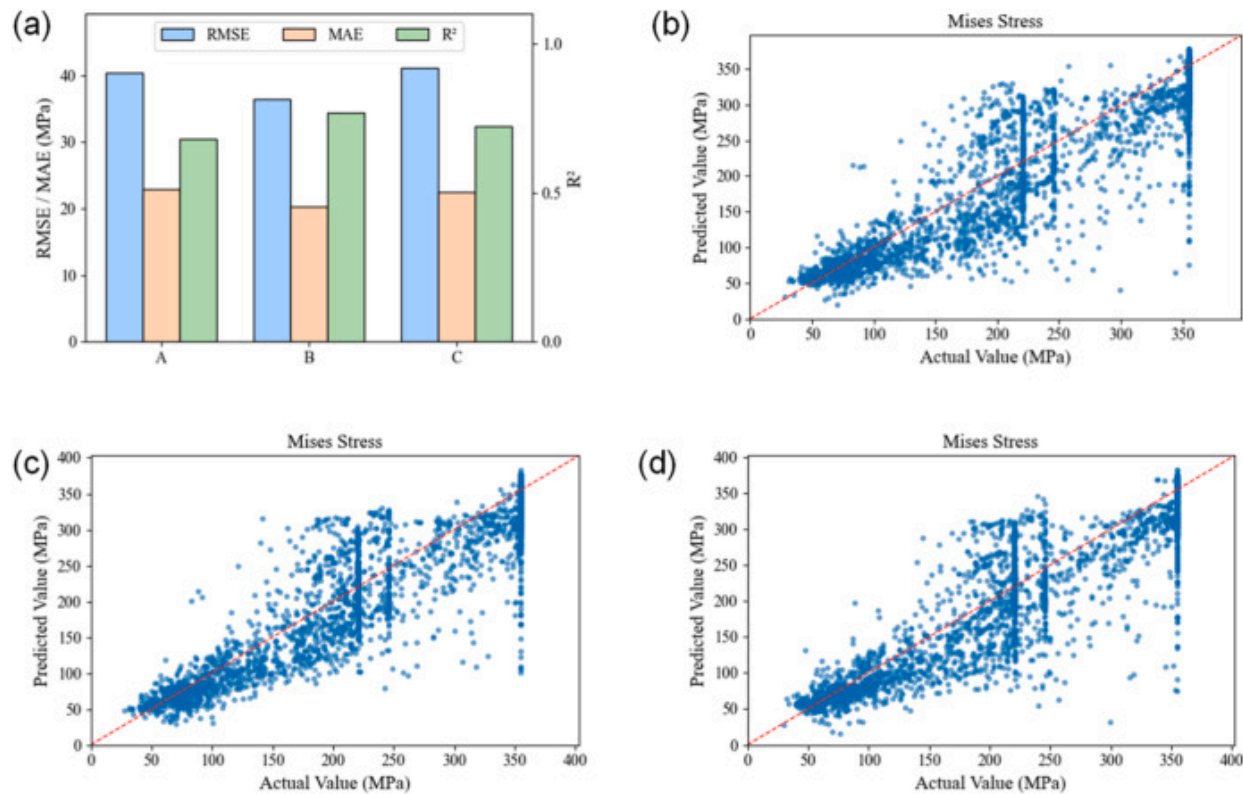
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Fig. 13. PINN sensitivity analysis: (a) physical coefficients, (b) number of hidden layers.

In order to evaluate the impact of network depth on prediction performance, the impact of the number of hidden layers (layers 1 to 6) on model performance was further analyzed. As shown in Fig. 14(b), the RMSE of the model first decreases and then increases as the number of layers increases. When the number of hidden layers is 4, the model achieves the best performance (RMSE=28.6MPa). Shallower networks (layers 1-3) fail to fully learn the complex nonlinear mapping of welding thermal-mechanical coupling due to insufficient expression capabilities, resulting in higher RMSE. An overly deep network (5-6 layers) may introduce over-parameterization problems, causing the training process to be unstable or fall into local optimality, thereby reducing

generalization ability. The four-layer hidden structure achieves a good balance between model expression ability and training stability, and its corresponding RMSE is significantly lower than other structures, which verifies the rationality of the current network architecture design.



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Fig. 14. Evaluation of ablation models: (a) overall performance metrics; comparison of predicted versus “true” von Mises stress for (b) Model A, (c) Model B, and (d) Model C.

Computational efficiency is a key indicator for evaluating the comparative value of PINN and FEM. Table 3 presents the calculation time of each model. Note that the calculation time analysis in this performed is as follows: the one-time offline time to generate the

training data set is about 4158s (including finite element simulation and data extraction preprocessing). It can be seen that the BPNN yields a shortest training time, but a poorest performance (Fig. 11(a)), and that of the PINN model is significantly shorter than that of PSO-BPNN (~40%), and the training efficiency is optimized while maintaining physical constraints. More importantly, after training is completed, PINN's online prediction of a single set of new working conditions only takes milliseconds, achieving an acceleration ratio of about four orders of magnitude compared to the equivalent FEM.

Table 3. Calculation time of each model.

Model	Time (s)
FEM	3613
BPNN	99
PSO-BPNN	763
PINN	295

4.4. Importance of physics-informed data inputs and constraints

To elucidate the sources of the model's performance improvement, ablation studies were conducted by sequentially removing two key design elements from the PINN: the maximum cooling rate and the physics-informed constraint term. This resulted in three comparative models:

Model A: uses only the three-dimensional node coordinates as input.

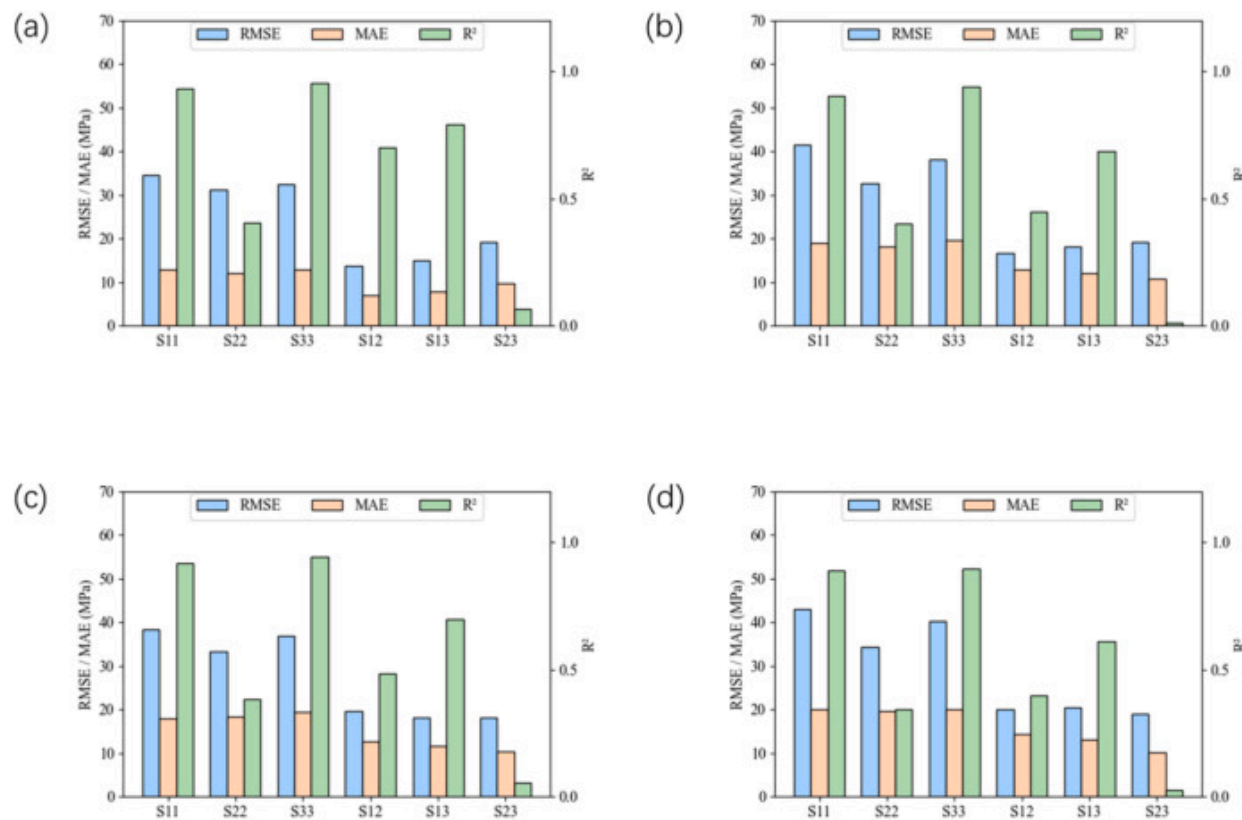
Model B: retains the nine-dimensional input but employs only the data-driven loss term.

Model C: uses the three-dimensional input with the data-driven loss term only.

As shown in Fig. 14, removing any key design element can cause a significant degradation in model performance. The complete PINN model outperforms all ablation variants in every evaluation metric: Model B's R^2 reaches 0.8603, Model A's is 0.8285, while Model C drops to 0.7228. Correspondingly, the RMSE of model B is 31.45 MPa and the MAE is 22.63 MPa. PINN has the lowest RMSE and MAE values among all models. These results highlight the complete input features of the maximum temperature of each welding stage, the three-dimensional coordinates of the nodes, the maximum cooling rate and the average cooling rate of the final cooling stage, as well as the physically based von Mises stress constraints, both of which play an important role in improving the prediction performance. The synergistic fusion of these multidimensional thermal inputs and physical regularization is the key to achieving accurate and reliable prediction of residual stress fields.

The influence of each ablation model on the prediction accuracy of each stress component is shown in Fig. 15. Across all six stress components, the full PINN achieved the highest or close to the highest R^2 values, demonstrating its robust multi-component prediction capabilities. When the thermal history enhancement input of three-dimensional coordinates is removed, corresponding to model A), the prediction accuracy of most stress components drops significantly. For example, the R^2 of S12 dropped from 0.7026 (PINN) to 0.4472, and that of S23 dropped significantly from 0.0671 to 0.0131, indicating that the model's ability to capture complex shear stress interactions was significantly weakened. When only the physical information loss term is removed and key inputs are retained (corresponding to model B), some components such as S11 and S33 still maintain high accuracy (R^2 are 0.9175 and 0.9450 respectively), but the predictive ability of the shear components (S12, S13, S23) decreases significantly, and the R^2 of S23 only reaches 0.0537. This shows that it is difficult for purely data-driven methods to learn the intrinsic physical correlation between the components of the stress tensor, and it is easy to overfit the main normal stress components. When the enhanced input and physical constraints are removed at the same time (corresponding to model C), the performance of the model further declines in each component. The R^2 of S23 drops to

0.0271 and that of S12 drops to 0.3996. These results jointly show that the maximum temperature, maximum cooling rate, and average cooling rate, as part of the thermal history input, provide key thermomechanical coupling information. More importantly, the physical information constraints effectively couple the learning process of all six stress components, enhance the consistency of the stress field, and improve the reliability of the overall prediction.



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Fig. 15. Prediction performance of individual stress components for different ablation models: (a) proposed PINNs, (b) Model A, (c) Model B, and (d) Model C.

Although the physical information neural network constructed in this study shows high accuracy and stability in predicting welding residual stress of dissimilar steels, there are still some limitations. Secondly, the proposed method is mainly suitable for welding joints similar to training conditions at the current stage. If it is applied to other welding scenarios with significant differences in welding methods, materials, geometry or process parameters, it is necessary to regenerate data and adjust the model. Finally, the von Mises criterion introduced into the model as a physical constraint improves the consistency of the stress field, but it fails to fully consider the complex physical-metallurgical behaviors such as dynamic phase transformation and creep that may occur during the welding process. Therefore, the prediction ability may be limited in joints involving significant microstructure transformation. Future research will further explore the transfer learning framework across process parameters and try to couple the constitutive relations of temperature and tissue stress fields to enhance the physical generalization ability and engineering applicability of the model.

5. Conclusion

This study addresses the need for high-fidelity and rapid prediction of residual stresses in dissimilar steel welds by developing a physics-constrained Physics-Informed Neural Network (PINN) framework. Experimental measurements, finite element simulations, and neural network modeling of a SUS304–Q235B welded joint were employed to systematically validate the proposed approach. The main findings are summarized as follows.

- (1) Comparison of experimentally measured thermal cycles with longitudinal residual stress distributions obtained via the contour method confirmed the reliability of the 3D thermo-elastic-plastic finite element model in simulating both the welding thermal process and the resulting residual stress field. This provides a high-quality, credible database for training the PINN.

- (2) The proposed PINN outperforms conventional BPNN and PSO-BP models, achieving superior prediction accuracy (RMSE=28.6MPa, MAE=9.06MPa, $R^2=0.9279$), training stability, and generalization capability.
- (3) Ablation studies demonstrate a significant synergistic effect between incorporating the maximum cooling rate and the physics-informed constraint. Their combined inclusion unifies thermo-mechanical process information with mechanics-based guidance, markedly improving predictions of secondary stress components, such as shear stresses.

CRediT authorship contribution statement

Huai Wang: Writing – review & editing, Writing – original draft, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Mengyao Wang:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Haoran Lin:** Validation, Investigation, Data curation. **Jin Wang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Qian Zhang:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Wenchun Jiang:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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